Name:_____Period:_____

DISCOVER ESSRT 1

The Mineral Identification Flowchart

Earth & Space Sciences Reference Tables, pages 16–17 · HS-ESS2-3

Directions: Follow the instructions to work through each part. Answer the questions in the spaces provided. Parts marked **PAPER** use only your printed Reference Tables. Parts marked **EXPLORER** use the Mineral Tests on the class website.

Vocabulary: cleavage, conchoidal fracture, fracture, hardness, luster, metallic, Mohs scale, nonmetallic, specific gravity, streak, streak plate, vitreous

Prior Knowledge Questions

Do these BEFORE you open your Reference Tables or the Explorer.

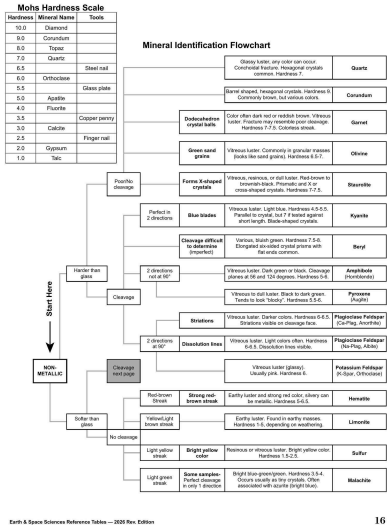
1. Real gold and “fool’s gold” (pyrite) are both shiny and yellow. Gold is very valuable; pyrite is not. If someone handed you a shiny yellow rock, what could you do to find out which one it is?

2. A bird guide tells you to check a bird’s size first, then its beak, then its colors. Why might it help to ask questions in a set order, instead of looking at everything at once?

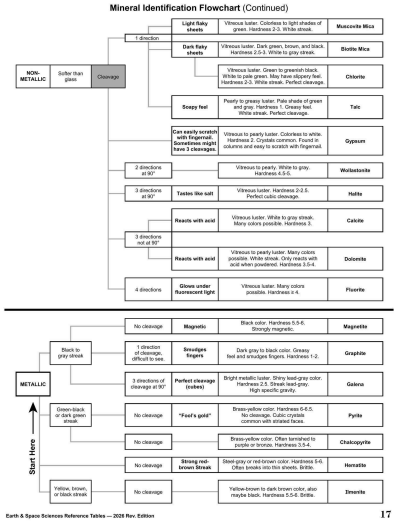
Warm-up

PAPER · No device needed

Open your printed Reference Tables to **pages 16 and 17**, the Mineral Identification Flowchart (shown below). The chart asks a series of questions about a mineral sample, and each answer sends you along a line to the next box. Before using the Explorer, get to know the pages.



ESSRT page 16 (NYSED, 2026 Rev. Edition)



ESSRT page 17

1. The chart has two places marked **Start Here**. For each one, write the page it is on and the word in the box it points to.

   Start Here #1: page _____ box: _____

   Start Here #2: page _____ box: _____

2. Find the **Mohs Hardness Scale** at the top left of page 16.

   A. Which mineral is the softest? Which is the hardest?

      Softest:_____

      Hardest:_____

   B. Which **tool** has a hardness of 5.5?

      Tool:_____

   C. List the four tools in order from softest to hardest.

      _____

3. On page 16, find the gray box that says **Cleavage next page**.

   A. Starting at Start Here, which two boxes come before it?

      _____

   B. At the top left of page 17, three boxes in a row repeat that path. What do they say?

      _____

4. Page 17 is split by a thick black line. Below it is the **metallic** side. How many mineral names are in its right-hand column?

   Number of metallic minerals:_____

5. One mineral's name appears in the right-hand column of **both** pages. Which one?

   Mineral:_____

Activity A — Test 1: Luster

Get the Explorer ready:	Question:
Open the **Mineral Tests** (pages 16–17) on the class website and click **Luster** (Test 1). Use **Next** and **Back** to move through the steps.	What is luster, and why is it the first question on the chart?

EXPLORER

1. Read step 1. What is **luster**? Where on the pages does each kind of mineral start?

   Luster is:_____

   Nonmetallic minerals start on page:____

   Metallic minerals start on page:____

2. **Observe:** Watch the light in the step 2 drawing. Describe what happens to the light at each kind of surface.

Surface	What happens to the light	How it looks
Metallic		
Nonmetallic: glassy		
Nonmetallic: earthy		

3. Read step 3. What question should you ask yourself to decide whether a luster is metallic? Name two colors a metallic mineral can be.

   _____

   _____

4. **Record:** Read step 4 and point at each photo. Complete the table of nonmetallic luster words.

Luster word	It looks like…	Example mineral in step 4
Vitreous		
Pearly		
Greasy		
Resinous		
Earthy (dull)		

5. **Analyze:** Read step 5. Pyrite and sulfur are both yellow.

   a. Which side of the chart does each start on? _____

   b. What does this tell you about using color to judge luster?_____

6. Read step 6.

   a. Which mineral appears on **both** sides of the chart?_____

   b. Why?_____

7. **Try it:** Do the practice round in step 7. Write your score, then explain why **muscovite mica** is nonmetallic even though it is shiny.

   Score: _____ of 8.

   Muscovite:_____

PAPER · Back to the page

8. On your printed page 17, read the description box in **galena's** row (just left of its name). Copy the words that tell you its luster. Then find **two** minerals on page 16 whose description says **vitreous**.

   Galena:_____

   Two vitreous minerals:_____&_____

Activity B — Test 2: Hardness

Get the Explorer ready:	**Question:**
Click **Finish** or **Close** to leave Test 1. Open the **Mineral Tests** (pages 16–17) on the class website and click **Hardness** (Test 2). Use **Next** and **Back** to move through the steps.	How do you test hardness, and what does "harder than glass" mean on the chart?

EXPLORER

1. Read step 1.
   a. After luster, what question does the **nonmetallic** side ask? _____

   b. What is the glass plate's hardness? _____

   c. Which side of the chart skips this question? _____

2. Read step 2. Complete the sentence: A _____ mineral scratches a _____ one, never the other way around.

   a. Diamond is 10 and corundum is 9. Is diamond only "a little" harder? Explain._____

   _____

3. **Record:** In step 3, run each scratch test. Write whether a scratch appears, and compare the two hardness numbers to explain why. Add one test of your own in the last row.

Tool	Mineral	Scratch?	Why (compare the numbers)
Fingernail	Calcite		

Tool	Mineral	Scratch?	Why (compare the numbers)
Copper penny	Calcite		
Steel nail	Quartz		
Glass plate	Quartz		
Glass plate	Fluorite		

a. In the glass plate test, what gets scratched: the mineral or the glass?_____

4. **Observe:** In step 4, rub calcite and quartz on the glass, then wipe.

a. What happened to each mark?_____

b. How can you tell powder from a real scratch?

5. Read step 5 and point at the photos.

a. At which numbered box do the paths split? _____

b. Which of the four minerals' paths continues on page 17?_____

6. **Explain:** In the step 6 practice, fluorite "scratches a copper penny, but a steel nail scratches it." Why is the answer **need another test**?

_____

_____

PAPER · Back to the page

Detail of page 16: Mohs Hardness Scale

7. A steel nail scratches a mineral, but a copper penny does not.

a. Its hardness is between which two numbers?

Between _____ and _____

b. Which minerals on the Mohs scale are in that range?

_____

c. Can you tell from these two tests whether it is harder than glass?

Explain._____

_____

Mohs Hardness Scale

Hardness	Mineral Name	Tools
10.0	Diamond	
9.0	Corundum	
8.0	Topaz	
7.0	Quartz	
6.5		Steel nail
6.0	Orthoclase	
5.5		Glass plate
5.0	Apatite	
4.0	Fluorite	
3.5		Copper penny
3.0	Calcite	
2.5		Finger nail
2.0	Gypsum	
1.0	Talc	

Activity C — Test 3: Cleavage & Fracture

Get the Explorer ready:	Question:
Open the **Mineral Tests** (pages 16–17) on the class website and click **Cleavage & fracture** (Test 3). In step 3, drag the shape to turn it, and use the **Split** buttons.	How does the way a mineral breaks help identify it?

EXPLORER

1. Read step 1. On page 16, what are the two cleavage choices on the **harder than glass** side? On the **softer than glass** side?

   Harder than glass:_____

   Softer than glass:_____

2. **Observe:** In step 2, tap each sample to break it. Describe each break, then define the two terms.

   Galena breaks:_____

   Quartz breaks:_____

   Cleavage is:_____

   Fracture is:_____

3. Point at the photos in step 5. Which mineral peels into thin sheets? Which breaks in **4** directions?

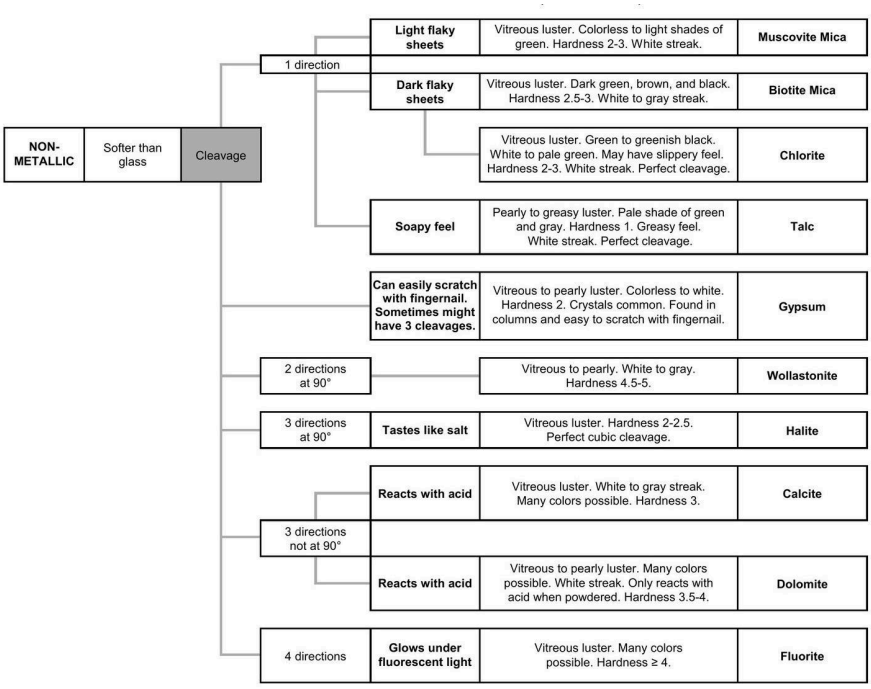
   Thin sheets:_____

   4 directions:_____

4. **Explain:** In the step 6 practice, pyrite has flat faces but the answer is **No cleavage**. Why?_____

_____

PAPER · Back to the page



Detail of page 17: the softer nonmetallic minerals with cleavage

7.A nonmetallic sample is softer than glass and breaks into perfect cubes.

a. After the **Cleavage** box on page 17, which box do you follow? _____

b. Which box comes next? _____

c. What is the mineral?_____

d. Two minerals are reached through **3 directions not at 90°**. Name both._____&_____

Activity D — Test 4: Streak

Get the Explorer ready:	Question:
Open the **Mineral Tests** (pages 16–17) on the class website and click **Streak** (Test 4). In steps 2 and 3, drag each sample across the white plate (or press **Rub it for me**).	Why is a mineral's streak more useful than its color?

EXPLORER

1.Read step 1. Where does the streak question come on the **nonmetallic** side? On the **metallic** side?

Nonmetallic:_____

Metallic:_____

2. **Record:** Read step 2. Streak each of the six samples and fill in the table, using the chart's words from the result cards.

Mineral	Color of the mineral	Color of its streak
Pyrite		
Galena		
Magnetite		
Sulfur		
Malachite		
Limonite		

a. What is a **streak plate**? _____

b. Which mineral's streak is the most different from its color?_____

3. **Analyze:** In step 3, streak both hematite samples.

a. What do you notice? _____

b. Why does the chart trust streak more than color?_____

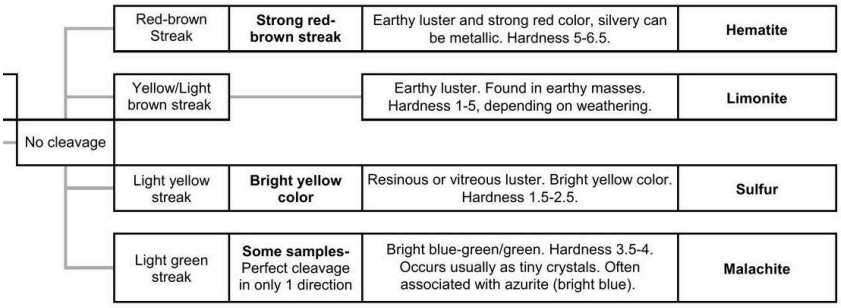
4. Read step 4. Hematite and limonite can both look brown. What streak color does each leave?

Hematite: _____ Limonite: _____

5. Read step 5. Magnetite and galena share the **black to gray streak** box.

a. What question does the chart ask next to tell them apart?_____

PAPER · Back to the page



Detail of page 16: softer nonmetallic minerals with no cleavage

6. A nonmetallic sample is softer than glass, has no cleavage, and leaves a **light green** streak.

a. What mineral is it?_____

b. Write one more clue printed in its row that you could check._____

Activity E — Test 5: Special Properties

Get the Explorer ready:	Question:
Open the **Mineral Tests** (pages 16–17) on the class website and click **Special properties** (Test 5). In steps 2–5, drag each tool onto the samples. The buttons under each bench do the same thing.	What final clues does the chart give in the last box of each row?

EXPLORER

1.Read step 1. What is the **last box** before each mineral's name called? Record the six special properties it lists.

Mineral	Special property (chart's words)
Talc	
Chlorite	
Graphite	
Magnetite	
Galena	
Dolomite	

2.**Record:** In step 2, rub the fingertip on each sample.

Sample	How it feels	What happened to the fingertip?
Talc		
Graphite		
Chlorite		
Quartz		

3.In step 3, bring the magnet near each sample.

a. Which one is pulled? _____

b. Why is the magnet a useful test for these four samples in particular?_____

_____

4.Read step 4.

a. What is **specific gravity**? _____

b. Put the two cubes on the balance. Which cube is heavier, and about how many times as heavy? _____

_____

5.**Record:** In step 5, test each sample with acid, then scratch off some powder with the nail and test the powder.

Sample	Acid on the sample	Acid on its powder
Calcite		
Dolomite		
Quartz		

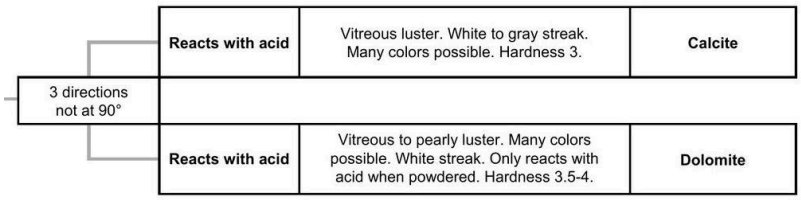
a. Explain what dolomite's box means by "Only reacts with acid when powdered."_____

_____

b. Write two safety rules for doing the acid test in a real lab._____

_____

PAPER · Back to the page



Detail of page 17: the two "Reacts with acid" rows

6.Two rows on page 17 have a **Reacts with acid** box.

a. Name both minerals. _____&_____

b. What printed in the description box tells them apart?_____

Put It Together — Identify the Sample

Get the Explorer ready:	**Question:**
Click ← **All mineral tests**. Click **Quiz: identify the sample** under the five tests.	Can you use all five tests together to name a mineral?

EXPLORER

1.**Try it:** Play a round of the quiz. For **two** of the samples, record the path you clicked and the name.

Sample	The path (box numbers and words)	Mineral	First try?
1			

Sample	The path (box numbers and words)	Mineral	First try?
2			

a. Which test did your hints mention most often? What will you check more carefully next time?_____

_____

PAPER · No device needed

3. Use only your printed pages 16 and 17. For each sample, follow the chart from Start Here and write every box you pass through, then the mineral's name.

Sample X

Test	Result
Luster	Metallic
Streak	Green-black
Breaks	No cleavage
Color	Brass-yellow, like gold
Hardness	6–6.5
Other	Cube-shaped crystals with fine lines on their faces

Path:_____

Mineral:_____

Sample Y

Test	Result
Luster	Nonmetallic, glassy
Hardness	Does not scratch glass (hardness 4)
Breaks	Cleavage in 4 directions
Color	Purple
Other	Glows under a fluorescent (UV) light

Path:_____

Mineral:_____

Sample Z

Test	Result
Luster	Nonmetallic, glassy
Hardness	Scratches glass (hardness 6–6.5)

Test	Result
Breaks	Cleavage in 2 directions at 90°
Color	Dark gray
Other	Fine parallel lines (striations) on its cleavage faces

Path:_____

Mineral:_____

Build It Better

Your Feedback on the Explorer

The Explorer is a **model** of your printed Reference Tables: the same data, shown a new way. Scientists and engineers test their models, find the strengths and limitations, and improve them. That's your job on this page. Your teacher is working with a company that builds the Explorer by coding, and feedback like yours is what tells it what to change in the next version.

No name needed.

Part of the Explorer: Mineral Tests and Mineral Quiz (pages 16–17)

Device: □ Chromebook □ Laptop □ Phone □ Tablet

1. **Rate it.** Circle one number for each.

	1 No	2	3	4 Yes!
It was easy to figure out what to click.	1	2	3	4
It helped me understand the printed page.	1	2	3	4
It worked well on the device I used.	1	2	3	4
I would use it again to study.	1	2	3	4

2. **Two models, one set of data.** Compare the Explorer with the printed page.

+ **Strength.** Something the Explorer shows **better** than the printed page:	– **Limitation.** Something the printed page does better, or something the Explorer **gets wrong or leaves out**:

3. **Fix one thing.** Be specific enough that someone could build it from what you write.

What kind?	□ Something broke □ Something was confusing □ An idea for something new
Where? Which step, screen, or button	
What happened? Or: what's missing	
What should happen instead?	

4. **Design it.** Sketch your change as it would look on the screen, and label it.

For teachers: NYSSLS Science & Engineering Practices: Developing and Using Models (evaluate the merits and limitations of a model) and Engineering Design (evaluate a solution against criteria; refine it). HS-ETS1-3.